

Analysing Business Processes, Alternatives and Operational Controls

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I developed an academic grocery-fulfilment business case and two supporting airport exercises. The work covers stakeholders, process and data models, timing assumptions, KPIs, redesign options, financial comparison, weighted evaluation and control design.

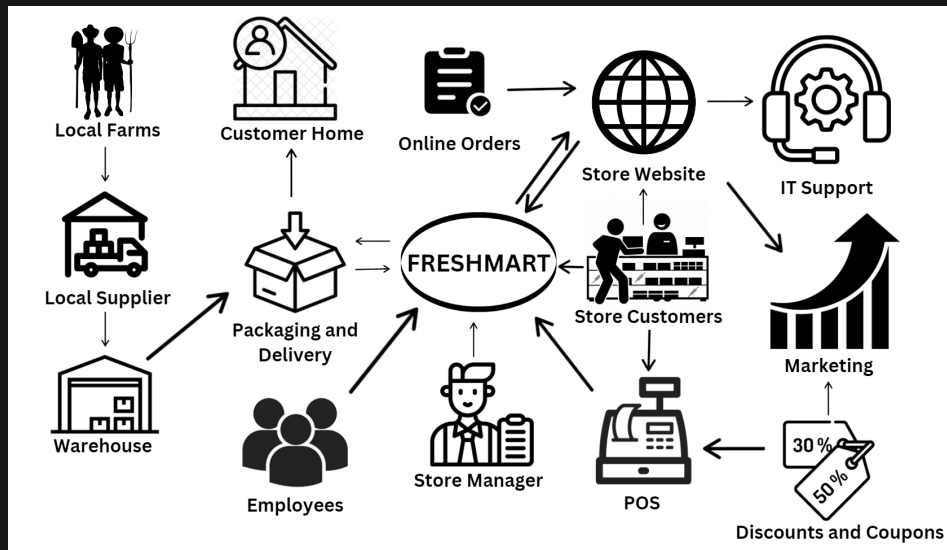
Tools and environment

Canva; Excel; Google Sheets; rich pictures; SIPOC; swimlanes; data-flow diagrams; KPI definitions; discounted cash-flow and weighted evaluation; control matrices.

Individual CITM410 work by Adnan Khan. June-July 2025. Business and airport details are case assumptions.

Identify the service context and process boundary

I used a rich picture and SIPOC to identify customers, suppliers, store operations, warehouse/delivery participants and technology support. The grocery case describes in-store and online sales. The detailed fulfilment boundary ends at courier handoff; travel to the customer is separate.



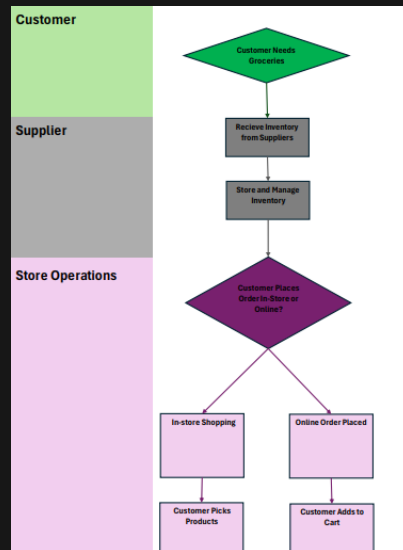
Stakeholder rich picture. Source portfolio p. 7.

Supplier	Input	Process	Output	Customer
Food Suppliers	Fresh Produce, Packaged Goods	1. Receive inventory from suppliers	Stocked Inventory	In-store Customers
Warehouse Providers	Storage space and equipment	2. Store and manage inventory	Online & In-store products	Online Customers
Delivery Partners	Delivery vehicles, drivers	3. Process customer orders (in-store & online)	Delivered groceries	Online Customers
Point-of-Sale System	Sales data	4. Handle checkout (cashier or online payment)	Sales receipts	All Customers
Customers	Orders (in-store & online)	5. Customer receives groceries (pickup/delivery)	Satisfied customers	Customer (end user)

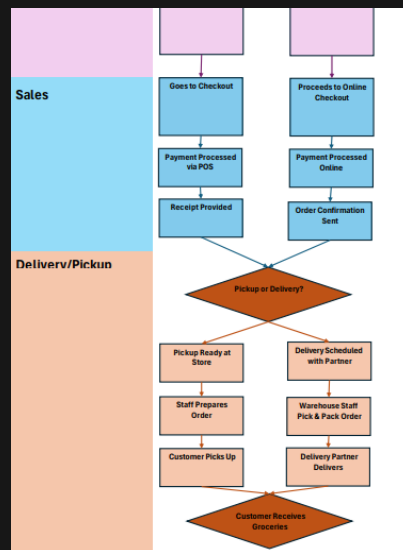
Sipoc process boundary. Source portfolio p. 8.

Map operating responsibilities and decisions

I built a high-level process map and swimlanes to follow order handling across operating roles. These views identify the sequence and the handoffs, including stock availability and preparation. They provide the baseline against which proposed changes can be reviewed.



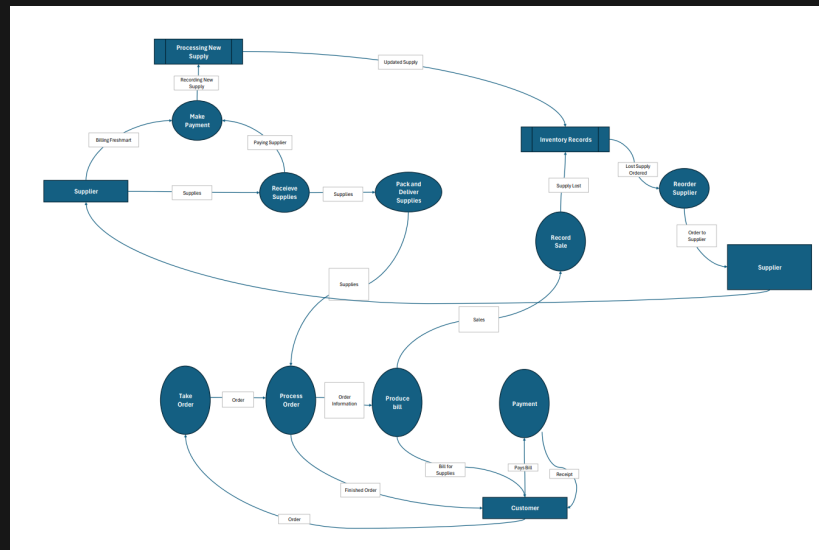
Roles and handoffs. Source portfolio p. 12.



Roles and handoffs. Source portfolio p. 13.

Trace information through the process

I developed a data-flow model for order, payment, inventory and supplier information. I also defined named flows by source, destination and purpose. A data-flow view is useful when the problem concerns a missing or incorrect record rather than only the order of tasks.



Data flow model. Source portfolio p. 14.

Data Element	Source	Destination	Description / Purpose
Order	Customer	Take Order	Customer places an order for goods/services
Order	Take Order	Process Order	Order details passed for processing
Finished Order	Process Order	Customer	Completed order delivered to customer
Bill for Supplies	Produce Bill	Customer	Invoice-bill sent to customer for payment
Pays Bill	Customer	Payment	Customer makes payment for goods/services
Receipt	Payment	Customer	Confirmation of payment sent to customer
Sales	Payment	Record Sale	Payment information recorded as a sale
Supplies	Supplier	Receive Supplies	Goods delivered from supplier to business
Supplies	Receive Supplies	Pack and Deliver Supplies	Supplies passed for packing and delivery
Supplies	Pack and Deliver Supplies	Record Sale	Supplies information passed for sales recording
Supplies	Pack and Deliver Supplies	Supplier	Unused or returned supplies, if applicable
Recording New Supply	Make Payment	Supply IT System	Update supply records after payment to supplier
Updated Supply	Supply IT System	Sales IT System	Updated inventory info sent to sales system
Supply Lost	Sales IT System	Reorder Supplier	Notification of lost/missing supply
Lost Supply Ordered	Reorder Supplier	Record Sale	Record of reordered supplies for sales tracking
Order to Supplier	Reorder Supplier	Supplier	New supply order sent to supplier
Billing Freshmart	Supplier	Make Payment	Invoice from supplier to business
Paying Supplier	Make Payment	Receive Supplies	Payment confirmation for received supplies

Data element definitions. Source portfolio p. 15.

Calculate normal and delayed fulfilment paths

The assumed normal path totals 33 minutes from order receipt to courier handoff. With 85% immediate availability and a 15% chance of an additional 2,880-minute delay, the expected time is 465 minutes under the additive-delay interpretation.

- Normal path: receive 2 + payment 3 + inventory 5 + pick 10 + package 6 + label 2 + handoff 5 = 33 minutes.
- Expected time: $0.85 \times 33 + 0.15 \times 2,913 = 465$ minutes.
- This is a calculation of case assumptions, not a time study or proof of two-hour delivery.

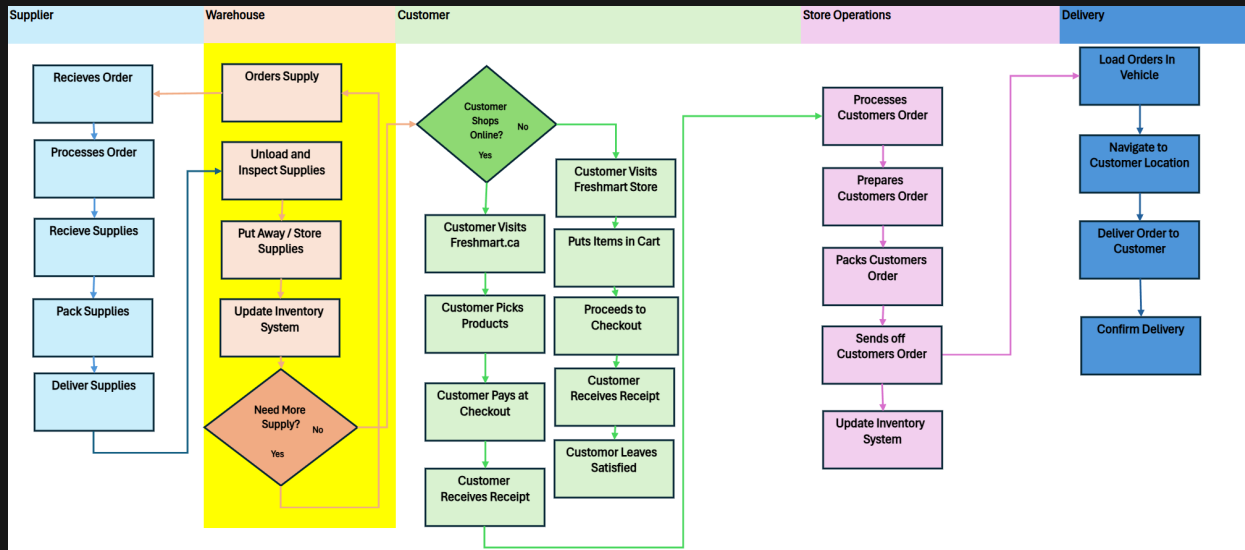
Define KPIs and how they would be measured

I separated order-to-handoff time, order accuracy, stock availability and inventory-record accuracy. These measures have different denominators and boundaries. The proposed review combines operational timestamps, service/return records and inventory information.

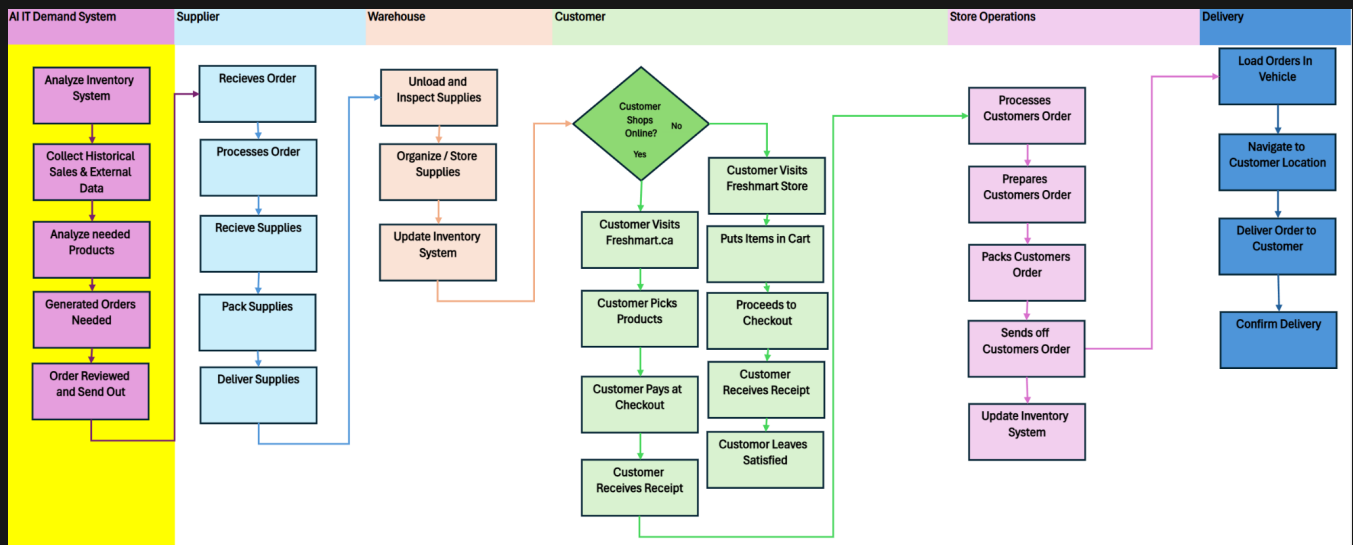
- Order accuracy: error-free shipped orders / all shipped orders.
- Availability: eligible orders fillable immediately / eligible orders.
- The control plan targets at least 90% accuracy and availability, with a separate 98% inventory-record accuracy target. These are proposed targets.

Compare the three operating alternatives

Option 1 retains the status quo. Option 2 clarifies responsibilities and handoffs while improving inventory visibility. Option 3 proposes extensive automation, including AI-assisted demand/reordering and automated fulfilment. The options differ in scope, cost and implementation demands.



Option 2 process redesign. Source portfolio p. 21.



Option 3 automation concept. Source portfolio p. 22.

Review the financial model on a consistent basis

The case uses five years and a 15% annual discount rate. The corrected full-precision comparison gives NPVs of \$335,215.51, \$554,305.22 and \$1,666,626.76 for Options 1, 2 and 3. These values preserve the source's Year 1 development-outlay timing.

- Development outlays: \$0, \$35,000 and \$500,000.
- Options 2 and 3 assume 5% annual growth in recurring costs and benefits.
- The ranking depends on estimated benefits and costs; it is not realized profit or a vendor-backed forecast.

Use weighted criteria to support the decision

I compared the options using seven criteria covering financial, operational and strategic considerations. The weighted scores are 23, 63.05 and 88.5. Option 3 ranks first under the chosen ratings and weights.

- Weights: NPV 25%; development cost 10%; cycle-time reduction 20%; bottlenecks 15%; work in process 10%; customer lead time 10%; competitive advantage 10%.

Criteria	% Weight	# Option 1: Status Quo (Baseline)	# Option 2: Process Redesign	# Option 3: Hyper Automated System
Financial Viability				
Net Present Value (NPV) (5 years)	25.00%	20	33	100
Initial Development Cost	10.00%	100	93	10
Operational Efficiency				
Average Cycle Time Reduction	20.00%	10	70	100
Bottleneck Identification	15.00%	20	70	90
WIP Management	10.00%	10	70	90
Strategic Advantage				
Enhanced Customer Lead Time	10.00%	10	70	100
Competitive Advantage	10.00%	10	70	100
Total Weighted Score	100.00%	23	63.05	88.5

Weighted evaluation matrix. Source portfolio p. 28.

Link proposed controls to accountable work

The Option 2 control plan includes seven points: order validation, inventory tracking, fulfilment procedures, payment security, courier handoff, label/information checks and KPI review. It also maps validity, completeness and accuracy objectives for information.

Recommended Control Plan	Control Goals of the Business Process (Option 2)																								
	Control Goals of the Operations Process							Control Goals of the Information Process																	
	Operations Process			For Input: Order Data	For Master Data: Inventory	For: Customer Service	Ensure Efficient Employment of Resources	Ensure Security of Resources	Order Input Validity			For Update Completeness		Inventory Input Validity											
	Ensure Effectiveness								IV			IC		IA		UC		UA		IV		IC		IA	
	A	B	C																						
P1: Order Receipt & Validation - Automated system validation - Customer credential verification - Payment method validation - 2-minute processing target	P1	P1		P1	P1	P1	P1	P1	P1	P1	P1			P1											
P2: Inventory Management - Real-time tracking system - Barcode scanning - Daily cycle counting - 98% accuracy target	P2	P2	P2		P2	P2	P2	P2				P2	P2	P2	P2	P2									
P3: Order Fulfillment Process - Defined process lanes - Standardized procedures - Performance dashboards - 33-minute cycle time	P3	P3	P3		P3	P3	P3					P3	P3												
P4: Payment Processing Security - PCI DSS compliance - Encrypted transmission - Multi-factor authentication - Zero fraud tolerance						P4	P4	P4				P4	P4												
P5: Delivery Handoff Verification - Scanning of shipping label at courier handoff - Automated system update of order status to 'Out for Delivery' - Digital signature/confirmation from courier	P5	P5				P5	P5	P5				P5	P5												
P6: Shipping Label Generation & Verification - Automated generation of shipping labels - Integration with courier systems for real-time data	P6	P6				P6	P6					P6	P6												

Freshmart control matrix. Source portfolio p. 31.

Labels	P6	P6				P6	P6					P6	P6			
- Integration with courier systems for real-time data -Barcode verification of label data against order details																
P7: Performance Monitoring & Continuous Improvement Review -Regular review of key performance indicators (KPIs) and dashboard data - Root cause analysis for performance deviations - Implementation and tracking of corrective and preventive actions	P7	P7	P7			P7	P7	P7				P7	P7			
Define Control Goals A B and C:																
A: Achieve and consistently maintain an average order fulfillment cycle time of 33 minutes or less, enabling granular analysis and continuous process optimization.																
B: Maintain an order accuracy rate of 90% or higher for all shipped orders, effectively minimizing errors and associated rework costs through improved process steps.																
C: Ensure an inventory availability rate of 90% or higher, actively leveraging clearer inventory management processes to reduce backorders and optimize stock flow.																
Legend:																
IV: Input Validity - Ensuring input data represents actual economic events																
IC: Input Completeness - Ensuring all valid events are captured																
IA: Input Accuracy - Ensuring events are correctly entered																
UC: Update Completeness - Ensuring all entries are reflected in master data																
UA: Update Accuracy - Ensuring data is correctly reflected in systems																

Freshmart control matrix. Source portfolio p. 32.

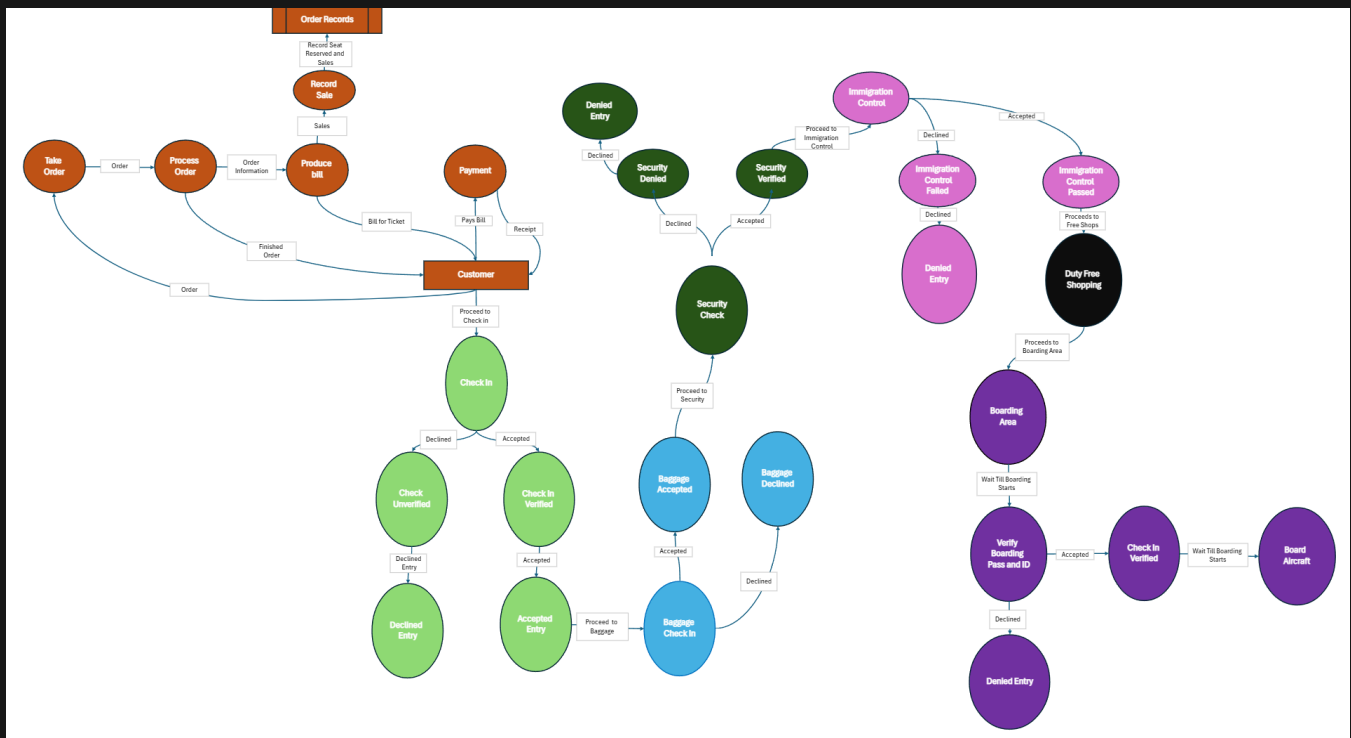
State the recommendation and implementation conditions

The academic case recommends Option 3 on its modelled NPV and weighted ranking. Before implementing that concept, the next work would validate data, cost estimates, integrations and exception paths, then test a limited pilot and controls tailored to the chosen option.

- The Option 2 control matrix is not automatically a complete control design for Option 3.
- Report realized outcomes separately from targets and modelled benefits.

Apply the method to the airport information-flow case

My airport exercise maps information across passenger, baggage and operating processes. I defined passenger cycle time, screening throughput and immigration-clearance lead time as separate measures. Each needs its own start/end points and evidence.



Airport data flow exercise. Source portfolio p. 35.

Extend the airport case into control design

The second airport exercise proposes controls for checkpoints, baggage, ground operations, emergency response, maintenance, reporting and feedback. The matrix links mechanisms to operational and information-quality objectives. These are coursework control proposals rather than deployed airport systems.

Recommended Control Plan	Control Goals of the Business Process (Option 2)																
	Control Goals of the Operations Process									Control Goals of the Information Process							
	Operations Process			For Input: Travel/Passenger Data	For Master Data: Operational Resources	For Stakeholder Coordination	Ensure Efficient Employment of Resources	Ensure Security of Resources	Operational Data Input Validity			For Update Completeness		Security Data Input Validity			
	Ensure Effectiveness								IV	IC	IA	UC	UA	IV	IC	IA	
	A	B	C														
P1: Integrated Smart Checkpoint Optimization System -Dynamic queue management and passenger routing via smart sensors and digital signage -Automated passport and boarding pass scanning with real-time data validation -Predictive analytics for security lane staffing and resource allocation -Centralized dashboard for real-time monitoring of passenger flow and checkpoint status	P1	P1	P1	P1	P1	P1	P1	P1	P1	P1	P1	P1	P1	P1	P1	P1	
P2: Intelligent Baggage Screening & Reconciliation System -Advanced X-ray/CT scanning with AI-powered threat detection -Automated baggage sorting and routing based on flight data -Real-time baggage tracking from check-in to aircraft loading -Automated reconciliation of bags with boarded passengers	P2	P2	P2	P2	P2	P2	P2	P2	P2	P2	P2	P2	P2	P2	P2	P2	
P3: Centralized Airfield Management & Ground Operations Control Tower -Real-time aircraft and ground vehicle tracking via radar/GPS -Integrated gate management and resource allocation module -Predictive analytics for optimizing taxiing and take-off/landing sequences -Centralized communication and incident response coordination for ground operations	P3	P3	P3	P3	P3	P3	P3	P3	P3	P3	P3	P3	P3	P3	P3	P3	
P4: Integrated Emergency Response & Crisis Management System -Centralized platform for real-time incident reporting -Automated alert dissemination to relevant agencies -Coordinated response protocols and communication channels -Post-incident analysis and reporting for continuous improvement	P4	P4				P4	P4	P4	P4	P4	P4	P4	P4	P4	P4	P4	
P5: Aircraft Maintenance & Safety Compliance Tracking System -Automated scheduling and tracking of mandatory aircraft inspections and maintenance -Digital logging of all maintenance activities and parts used -Real-time alerts for overdue maintenance or non-compliance issues -Centralized repository for safety certifications and regulatory documentation	P5	P5				P5		P5	P5	P5	P5	P5	P5				
P6: Regulatory Compliance Reporting & Archiving System -Automated compilation of operational data for regulatory reports (e.g., flight delays, security incidents, environmental metrics). -Secure archiving of all compliance documentation. -Automated submission reminders for regulatory deadlines. -Centralized audit trail for regulatory submissions.	P6	P6				P6	P6	P6				P6	P6				
P7: Customer Feedback & Satisfaction Survey System -Automated distribution of post-experience surveys to passengers -Secure collection and aggregation of survey responses -Automated sentiment analysis and trend reporting on customer feedback -Integration with customer service dashboards for performance review	P7					P7						P7	P7				
Define Control Goals A, B and C:																	
A: Achieve and consistently maintain an average passenger processing time of 15 minutes or less from curb-to-gate for departing passengers, enabling efficient throughput and minimizing delays.																	
B: Guarantee 100% comprehensive security screening and compliance for all passengers, baggage, and cargo, preventing unauthorized access or threats to aviation safety.																	
C: Ensure real-time, accurate, and integrated operational data across all critical airport systems, facilitating proactive decision-making and seamless inter-agency coordination.																	
Legend: IV: Input Validity - Ensuring input data represents actual economic events IC: Input Completeness - Ensuring all valid events are captured IA: Input Accuracy - Ensuring events are correctly entered UC: Update Completeness - Ensuring all entries are reflected in master data UA: Update Accuracy - Ensuring data is correctly reflected in systems																	

Airport control matrix. Source portfolio p. 37.

What I learned

- Define the process boundary before interpreting a cycle-time result.
- A ranking is only as useful as its assumptions and consistent calculations.
- A control plan needs ownership, evidence and a response when a target is missed.

Source: the full CITM410 portfolio supplied for this project. Evidence is from the original coursework; this edition adds no new lab test results. Repeated captures and earlier drafts are consolidated. Revised design summaries are labelled, and sensitive identifiers are masked.